Local Green, Global Red III

A finite evaluator overlay across six system roles

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Abstract

Two checks can pass separately while their composition fails. For binary local verdicts e_1, e_2 and a binary verdict e_g on the composite, the signed evaluator defect

$$\Delta_E = e_g - e_1 e_2$$

is -1 precisely in the local-green/global-red case $(e_1, e_2, e_g) = (1, 1, 0)$. This paper shows how the same finite diagnostic can be used across six distinct system roles without treating it as a new layer or changing its mathematics. The main results are elementary but useful: the defect has a three-way sign classification; finite sums obey an exact global-minus-local identity; cancellation of a total defect does not imply pointwise agreement; and an evaluator cannot descend to a quotient if it assigns different verdicts to equivalent carrier elements. A tested Python implementation, seven teaching and verification notebooks, and a Lean 4 library accompany the paper. The Lean library checks the finite integer statements and the quotient obstruction. The real-valued weighted-kernel theorem remains conditional on the explicit hypotheses inherited from LGGR I [8].

1. Why this matters

A composed system is often inspected by several specialists: one checks the mathematical carrier, another checks representation transport, another fixes the observation scale, and another evaluates an integrated run. Their jobs are different, but each can ask the same concrete question:

> Did the verdict on the composite agree with the product of the verdicts on its two parts?

Writing that discrepancy down prevents a locally successful pipeline from being summarized by a single misleading green status. The overlay is small enough to replay and strict enough to reject malformed inputs.

2. Scope

The results concern finite binary evaluations and declared integer or real weights. They do not establish reliability for an arbitrary agent, organization, software stack, or physical system. A zero observed defect is sample-relative and can also be produced by an empty sample or an uninformative evaluator. The real exponential kernel is proved here only conditionally in prose, not fully mechanized in Lean. The six-role interpretation is a recording convention, not an empirical adequacy result, and the related-work comparisons do not identify this model with another theory. These are the

standing caveats for every result below.

3. Finite evaluator calculus

3.1 Verdict triples

An evaluated boundary consists of a triple

$$v = (e_1, e_2, e_g) \in \{0, 1\}^3,$$

where e_1 and e_2 are local verdicts and e_g is the verdict after composition. Define

$$\Delta_E(v) = e_g - e_1 e_2.$$

Because $e_1 e_2, e_g \in \{0, 1\}$, the defect lies in $\{-1, 0, 1\}$.

3.2 Sign classification

Proposition 1. Every verdict triple has defect in $\{-1, 0, 1\}$:

$$\begin{aligned} \Delta_E = -1 &: \text{ local green, global red,} \\ \Delta_E = 0 &: \text{ the composite verdict equals the local product,} \\ \Delta_E = 1 &: \text{ local product red, global verdict green.} \end{aligned}$$

Proof. Both e_g and $e_1 e_2$ are binary, so their difference has one of the three displayed values. □

The signs are retained. Replacing Δ_E by $|\Delta_E|$ is useful for magnitude summaries but loses the direction of disagreement.

3.3 A one-trace counterexample

Proposition 2. Local success does not imply composite success.

Proof. Choose $(e_1, e_2, e_g) = (1, 1, 0)$. Then both local verdicts pass, the composite verdict fails, and

$$\Delta_E = 0 - (1)(1) = -1. \qquad \square$$

This is an existence result. It says that compositional success needs additional hypotheses; it does not estimate how often the failure occurs.

4. From one trace to a finite sample

Let $v_i = (e_{1i}, e_{2i}, e_{gi})$ for $i = 1, \ldots, n$ and define

$$G = \sum_{i=1}^{n} e_{gi}, \qquad L = \sum_{i=1}^{n} e_{1i} e_{2i}, \qquad D = \sum_{i=1}^{n} \Delta_E(v_i).$$

Theorem 1. For every finite sample,

$$G - L = D.$$

Proof. Substitute the definition of Δ_E and use finite distributivity:

$$\sum_i (e_{gi} - e_{1i}e_{2i}) = \sum_i e_{gi} - \sum_i e_{1i}e_{2i}. \qquad \square$$

4.1 Cancellation and disagreement count

Proposition 3. $D = 0$ does not imply $\Delta_E(v_i) = 0$ for every i.

Proof. The two triples $(0, 1, 1)$ and $(1, 1, 0)$ have defects 1 and -1. Their sum is zero although neither defect is zero. $\square$

To distinguish signed cancellation from total agreement, define the positive and negative defect counts:

$$N_+ = \#\{i : \Delta_E(v_i) = 1\}, \qquad N_- = \#\{i : \Delta_E(v_i) = -1\}.$$

Then the total signed defect and the total disagreement count satisfy:

$$D = N_+ - N_-, \qquad M = N_+ + N_- = \sum_{i=1}^{n} |\Delta_E(v_i)|.$$

While $D = 0$ allows interior cancellations ($N_+ = N_- > 0$), $M = 0$ is strictly equivalent to pointwise agreement $\Delta_E(v_i) = 0$ on every item of the observed sample. For nonnegative weights $w_i \geq 0$, the weighted sum $\sum_i w_i |\Delta_E(v_i)| = 0$ certifies agreement on the positive-weight support. This is why the implementation reports both the signed defect sum D and the disagreement count M, alongside the pointwise defect list.

4.2 Worked five-trace example

Consider:

trace	e_1	e_2	e_g	Δ_E
1	1	1	1	0
2	1	1	0	-1
3	1	1	0	-1
4	0	1	0	0
5	0	0	0	0

The totals are $G = 1$, $L = 3$, and $D = -2$, hence $1 - 3 = -2$. The companion Python test and Lean theorem compute these same integers.

5. Evaluated path kernels

Let A, B, C be finite state sets. For a role profile r, let $\Gamma_r(b, a)$ be the finite set of admitted paths from a to b, with cost $C_r(\gamma)$ and evaluator $E_r(\gamma) \in \{0, 1\}$. For $\lambda \geq 0$, define

$$Z_r(b \mid a) = \sum_{\gamma \in \Gamma_r(b,a)} E_r(\gamma) \exp[-\lambda C_r(\gamma)].$$

For proposal-weighted paths, an additional nonnegative proposal mass is multiplied into each summand.

The composition hypotheses are:

- **G1:** composition is a bijection

$$\bigsqcup_{b \in B} \Gamma_r(c, b) \times \Gamma_r(b, a) \longrightarrow \Gamma_r(c, a).$$

  Thus every admitted composite has one compatible segment pair, and every compatible pair produces exactly one admitted composite;
- **G2:** cost is additive across that decomposition;
- **G3:** the evaluator factorizes across the boundary;
- **G4:** proposal mass factorizes across the boundary when proposal weights are used.

Theorem 2 (conditional on G1–G4). Under G1–G3, and also G4 in the proposal-weighted case,

$$Z_r(c \mid a) = \sum_{b \in B} Z_r(c \mid b) Z_r(b \mid a).$$

Proof. Reindex the finite sum over $\Gamma_r(c, a)$ along the G1 bijection. It becomes a sum over the displayed disjoint union of all compatible segment pairs. G2 replaces each composite cost by the sum of its segment costs, so the exponential is a product. G3 replaces each composite verdict by the product of its segment verdicts, and G4 does the same for proposal mass when present. Finite distributivity then turns, for each b, the sum over the Cartesian product into $Z_r(c \mid b) Z_r(b \mid a)$. Summing those products over b gives the displayed identity. □

When G3 is not assumed, the composite summand retains $E_r(\gamma_2 \circ \gamma_1)$ while the product of segment partition functions produces $E_r(\gamma_2) E_r(\gamma_1)$. Under G1, G2, and G4 when proposal weights are used, the same reindexing argument yields the exact discrepancy identity:

$$Z_r(c \mid a) - \sum_{b \in B} Z_r(c \mid b) Z_r(b \mid a) = \sum_{\text{compatible pairs}} w_r(\gamma_2, \gamma_1) \left[E_r(\gamma_2 \circ \gamma_1) - E_r(\gamma_2) E_r(\gamma_1) \right],$$

where $w_r(\gamma_2, \gamma_1) = \exp[-\lambda(C_r(\gamma_2) + C_r(\gamma_1))]$ (multiplied by proposal masses when applicable). The evaluated partition-function discrepancy is thus directly the weighted sum of local evaluator defects $\Delta_E(\gamma_2, \gamma_1)$.

The Lean companion formally mechanizes the integer reindexing. If the composite weight list equals the flattened products of the boundary weight lists, the composite sum equals the sum

over boundaries of left-sum times right-sum, and the residual is zero. The Python implementation computes the G1 bijection and G2–G4 from concrete finite path records and checks that integer identity on integer lists. Real exponentials remain outside both mechanizations.

6. Quotient descent and carrier mixing

Let X be a finite carrier with equivalence relation $\sim$, quotient map $q : X \to X/\!\sim$, and evaluator $E : X \to \{0, 1\}$.

Theorem 3. A function $\bar{E} : X/\!\sim \to \{0, 1\}$ satisfying $\bar{E}(q(x)) = E(x)$ for every $x \in X$ exists if and only if

$$x \sim y \Longrightarrow E(x) = E(y).$$

Moreover, when such $\bar{E}$ exists, it is unique.

Proof. ($\Longrightarrow$) If $x \sim y$, then $q(x) = q(y)$. Applying $\bar{E}$ gives $E(x) = \bar{E}(q(x)) = \bar{E}(q(y)) = E(y)$. ($\Longleftarrow$) Define $\bar{E}([x]) = E(x)$ for each equivalence class $[x] \in X/\!\sim$. The condition $x \sim y \implies E(x) = E(y)$ ensures $\bar{E}$ is well-defined. By construction, $\bar{E}(q(x)) = E(x)$ for all $x \in X$. Uniqueness follows from the surjectivity of the quotient projection q: if $\bar{E}'$ also satisfies $\bar{E}'(q(x)) = E(x)$ for all x, then for every $[x] \in X/\!\sim$, $\bar{E}'([x]) = \bar{E}'(q(x)) = E(x) = \bar{E}([x])$. □

Corollary 3.1. If equivalent elements receive different verdicts, the evaluator does not descend to the quotient.

The Lean companion formally mechanizes this obstruction and includes an explicit inhabited witness, so the hypotheses are not vacuous.

7. The six-role overlay

The overlay attaches the same record

$$(r,\ e_1,\ e_2,\ e_g,\ \Delta_E,\ \text{context})$$

to six role profiles. The role changes the context fields that must be declared; it does not change the defect formula.

role profile	question asked before comparing verdicts
finite obstruction	Are carrier, degree, comparison map, quotient gauge, and replay data declared?
foundations	Is this a diagnostic record rather than a decision record?
representation	Do the operation and transport witness match?
observer	Were both scores computed in the same observation envelope?
continuum shadow	Is a finite calculation being kept distinct from a limit statement?
action and alignment	Is residual feedback routed as feedback rather than relabeled as success?

These questions specify what context must be recorded. They do not add a seventh role and do not turn a declared field into evidence that a real evaluator is adequate.

7.1 Observation envelope

Proposition 4. For the declared score $S_w(R) = wR$, changing w can change the score of the same residual.

Proof. Take $R = 1$, $w = 1$, and $w' = 2$. Then $S_w(R) = 1 \neq 2 = S_{w'}(R)$. □

Scores from different envelopes should therefore not be compared without an explicit transport rule.

7.2 Integration does not erase a defect

Proposition 5 (fixed observation). If integration leaves (e_1, e_2, e_g) unchanged, it leaves Δ_E unchanged.

Proof. Δ_E is a function of that triple. Substituting the same arguments gives the same value. □

8. What is recorded

For each finite observation the package records the two local verdicts, the composite verdict, their signed difference, the role in which the comparison was made, and enough context to tell whether the compared quantities share a carrier, transport rule, and observation scale. A zero says only that the composite verdict equals the product of the local verdicts for that record. When pointwise defects cancel, the pointwise list is retained. When an evaluator varies inside a declared equivalence class, quotient evaluation is refused. This paper defines no repair procedure or cochain complex.

9. Reproducible implementation

The package contains:

- a Python reference implementation of the finite records and guards;
- tests for valid inputs, invalid inputs, and twelve break-and-recover mutations;
- seven teaching and verification notebooks, each with plots and optional interactive exploration;
- a Lean 4 library for the integer identities and quotient statements, plus a current theorem-level axiom report;
- deterministic scripts for figures, the manuscript build, checksums, and the release archive.

An earlier omnibus snapshot is retained outside the public deposit for provenance. It is excluded from the archive and from current claim evidence.

10. Falsifiability

The model is directly testable. Any of the following would refute a stated result or expose an implementation mismatch:

1. a binary verdict triple produces a defect outside $\{-1, 0, 1\}$;
2. a finite sample violates $G - L = D$;
3. the implementation accepts a non-binary verdict;
4. a quotient evaluator is produced despite two equivalent elements receiving different verdicts;
5. the implementation reports G1, G2, G3, or G4 as satisfied although the concrete path records

violate the corresponding condition; or
6. a notebook result cannot be reproduced from the archived environment.

A failed hypothesis places the record outside the theorem's applicable case; it does not refute the conditional theorem. Mathematical counterexamples, implementation discrepancies, and environmental reproducibility failures are distinct outcomes.

The executable constant-output check is only a basic falsifier; the stronger task-specific adequacy obligation stated in Section 2 remains open.

11. Related work

Weighted automata provide the standard algebraic setting for finite paths with semiring-valued weights. Kleisli's construction and Moggi's computational use of monads explain why effectful composition is naturally expressed through Kleisli arrows. The sheaf-theoretic account of contextuality by Abramsky and Brandenburger is a nearby local-to-global obstruction language, but this paper does not identify its binary evaluator defect with contextuality or sheaf cohomology. Contract-based design supplies a neighboring language for assumption-sensitive composition. Recent cellular-sheaf work on distributed multi-agent coordination and system identification studies richer dynamical and topological settings; the present contribution is only a finite, replayable evaluator overlay.

12. Conclusion

The useful object is the signed discrepancy $\Delta_E = e_g - e_1 e_2$. It survives changes of role name, exposes local-green/global-red traces, adds linearly over finite samples, and refuses quotient descent when an evaluator varies inside an equivalence class. Everything else in the overlay is context: which carrier, transport, observation envelope, and replay data make a comparison meaningful.

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